

Kafka Topic Replication Issue SOP

Objective

Restore Kafka topic replication health when replication factor is not maintained or partitions become under-replicated.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T043
Issue	Kafka topic replication issue
Category	Streaming
Service	Kafka
Severity	High
Status	In Progress
Description	Replication factor not maintained
Expected Resolution	Rebalance partitions and brokers

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm under-replicated partitions, offline replicas, or replication factor mismatch.
2. Identify affected topics, partitions, leaders, and in-sync replicas.
3. Check broker availability, disk usage, and network health.
4. Review recent broker restarts, partition reassignment, or topic creation changes.
5. Run partition reassignment or preferred leader election only after approval.
6. Monitor ISR recovery and replication lag.
7. Validate replication factor matches production standard.
8. Update ticket with affected topic list and recovery evidence.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Under-replicated partitions	Broker unavailable	Restore broker and monitor ISR
Replication factor mismatch	Topic created with wrong config	Increase replication factor through reassignment
Offline replicas	Disk or network issue	Fix broker infra
Leader imbalance	Uneven broker load	Run preferred leader election if approved

Best Practices

1. Use approved replication factor for production topics.
2. Monitor ISR shrink and offline partitions.
3. Avoid creating production topics with replication factor one.
4. Schedule reassignment during low traffic windows.

References

[Apache Kafka Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [Kafka CLI Runbook](#)